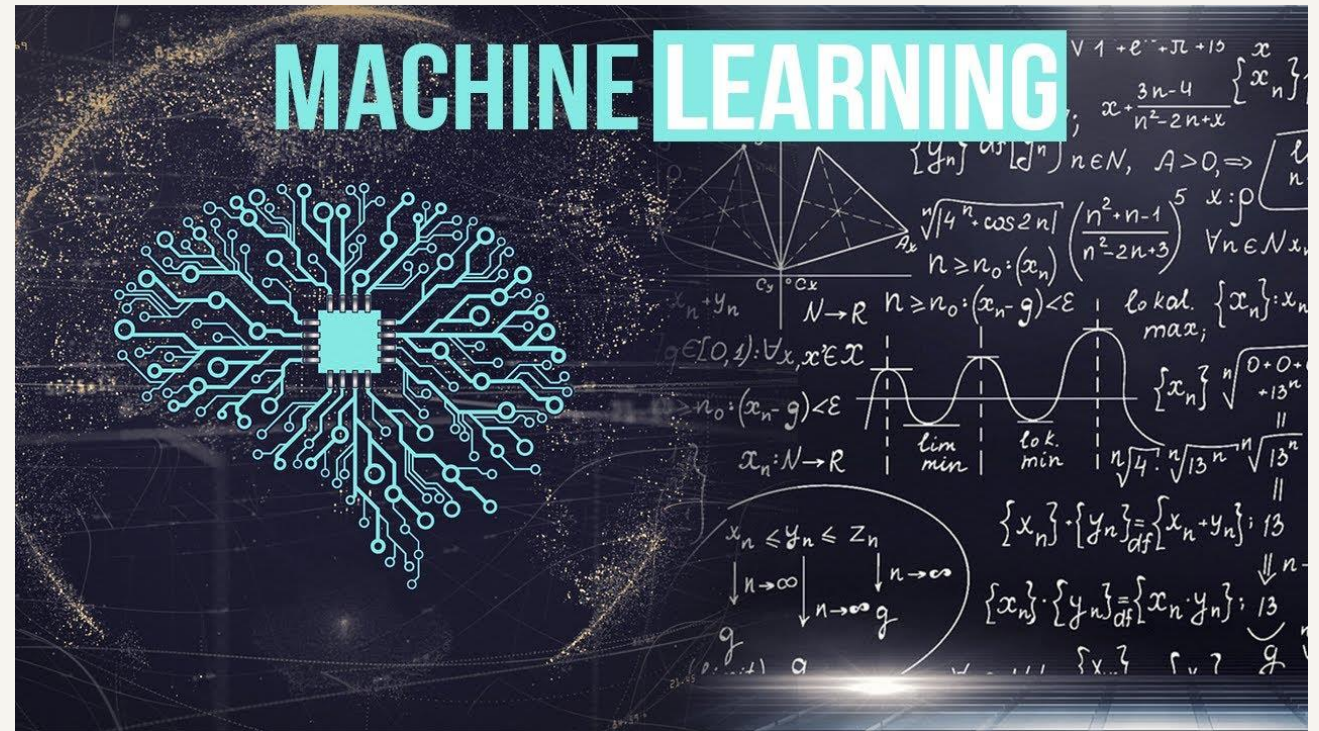


Introduction of Machine Learning



Index

- What is machine learning?
- Why and Where is machine learning?
- History of machine learning.
- Very first use case of machine learning.
- Human brain vs Machine

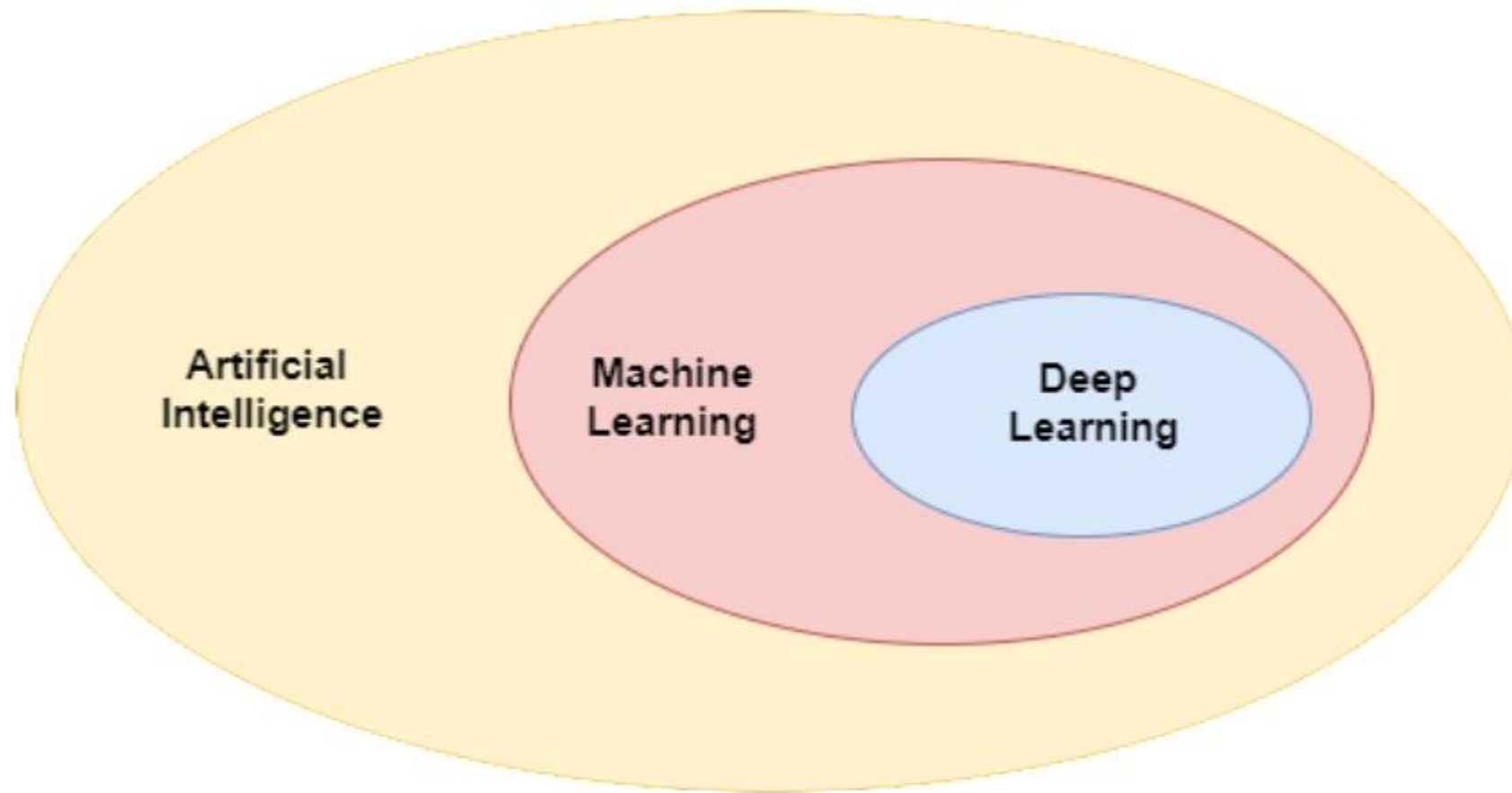
What is machine learning?

- Machine learning is a method of data analysis that uses algorithms that **iteratively learn from data** to find **hidden insights/patterns** automatically **without being explicitly programmed** where to look.

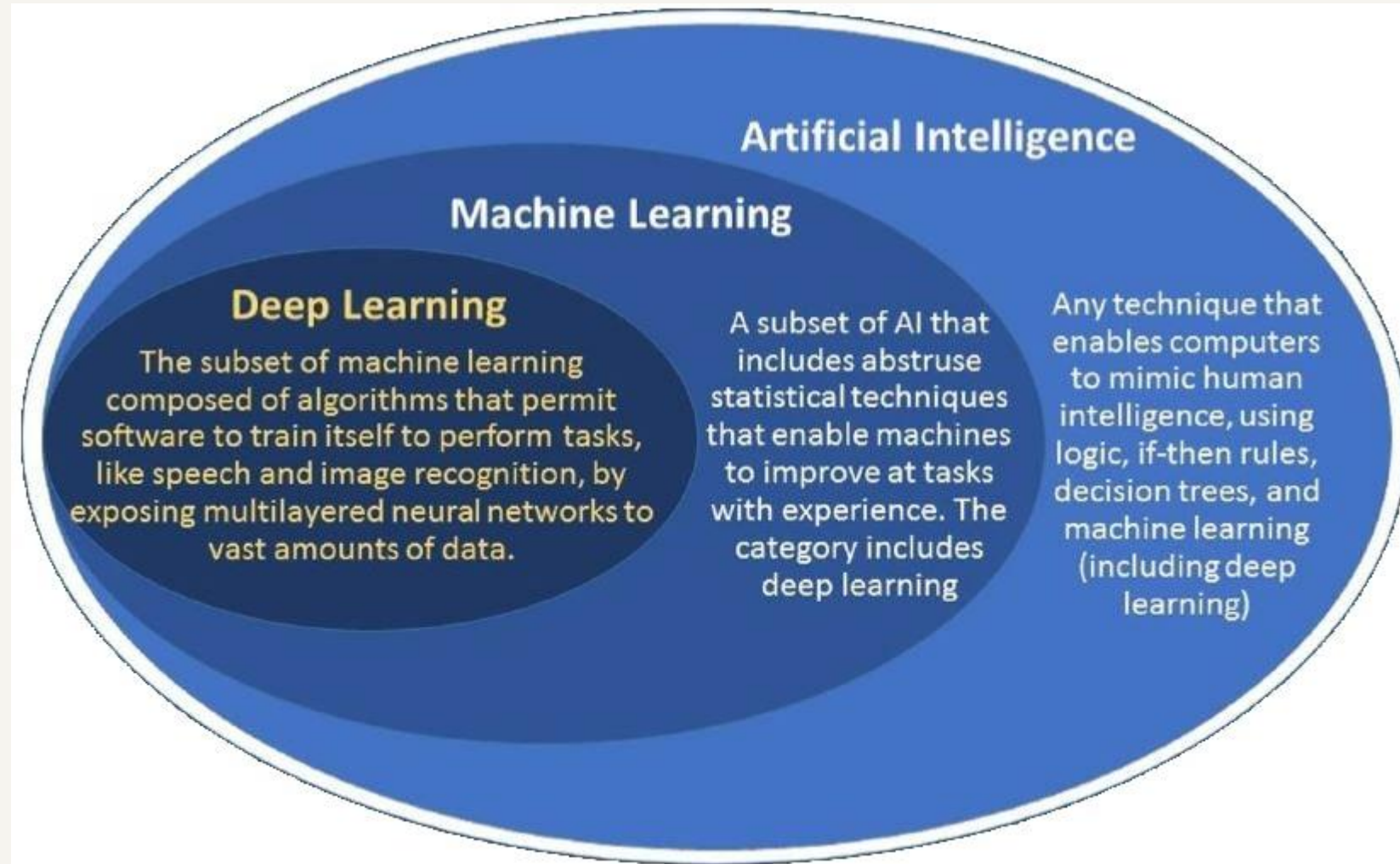
What is machine learning?

- Each task requires a different set of algorithms, and these algorithms **detect patterns** to perform certain tasks.
- Like Human learning from past experiences.
- A computer systems learns from **historical data**, which represents some **past experiences of an application domain**.
- The goal is to learn a **target function** that can be used to predict the response variable (Regression/Classification).

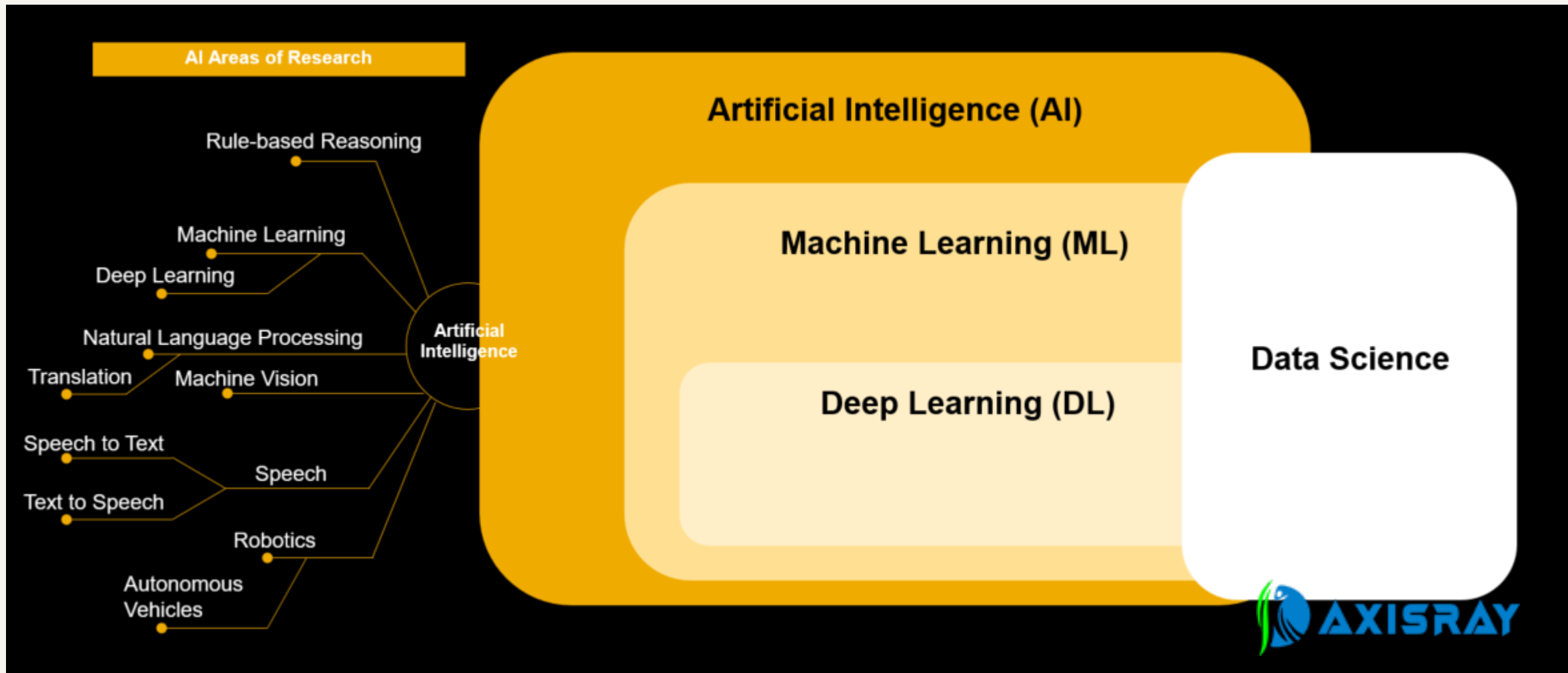
What is machine learning?



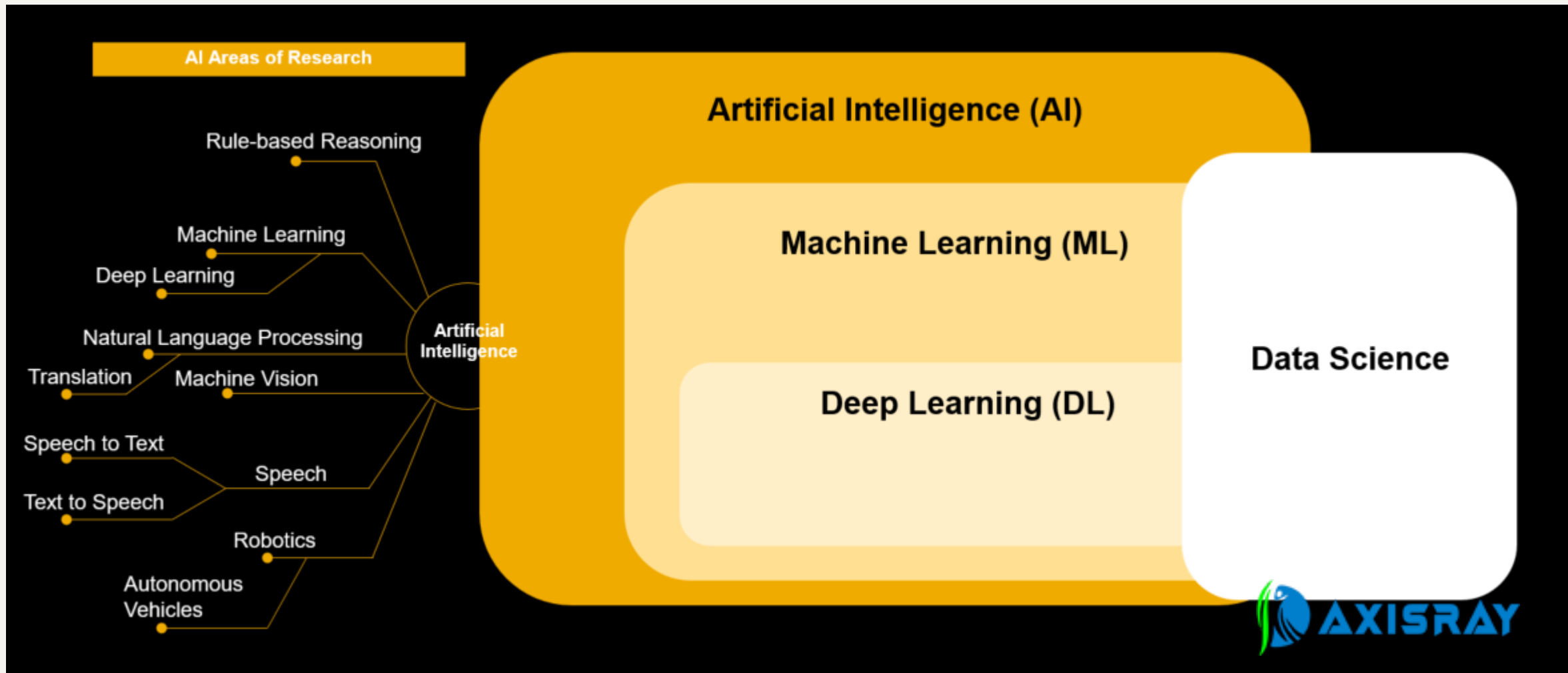
What is machine learning?



What is machine learning?



What is machine learning?



What is machine learning?



Why and Where is machine learning?

- If I ask you if you have ever utilized machine learning?

Why and Where is machine learning?

- When you ask Siri what the weather forecast is, that's machine learning.
- When you Google search something at work to help you do your job better or more efficiently, you can thank machine learning.
- Another everyday example is our spam folders—a machine learning algorithm is used to determine which emails are inbox-worthy, and which are spam and don't deserve attention.
- Similarly, when Netflix suggests a show, you should watch based on preference, it's getting the suggestion from an algorithm.

Why and Where is machine learning?

Financial services

Banks and other businesses in the financial industry use machine learning technology for two key purposes: to identify important insights in data and prevent fraud. The insights can identify investment opportunities, or help investors know when to trade. Data mining can also identify clients with high-risk profiles or use cybersurveillance to pinpoint warning signs of fraud.

Health care

Machine learning is a fast-growing trend in the health care industry, thanks to the advent of wearable devices and sensors that can use data to assess a patient's health in real time. The technology can also help medical experts analyze data to identify trends or red flags that may lead to improved diagnoses and treatment.

Why and Where is machine learning?

Government

Government agencies such as public safety and utilities have a particular need for machine learning since they have multiple sources of data that can be mined for insights. Analyzing sensor data, for example, identifies ways to increase efficiency and save money. Machine learning can also help detect [fraud and minimize identity theft](#).

Retail

Websites recommending items you might like based on previous purchases are using machine learning to analyze your buying history. Retailers rely on machine learning to capture data, analyze it and use it to personalize a shopping experience, implement a [marketing campaign](#), [price optimization](#), merchandise [supply planning](#), and for [customer insights](#).

Why and Where is machine learning?

Transportation

Analyzing data to identify patterns and trends is key to the transportation industry, which relies on making routes more efficient and predicting potential problems to increase profitability. The data analysis and modeling aspects of machine learning are important tools to delivery companies, public transportation and other transportation organizations.

Oil and gas

Finding new energy sources. Analyzing minerals in the ground. Predicting refinery sensor failure. Streamlining oil distribution to make it more efficient and cost-effective. The number of machine learning use cases for this industry is vast – and still expanding.

Why and Where is machine learning?



ANY
QUESTIONS
?

Deep Neural N
(Pretraini

SVM

Multi-layered
Perceptron
(Backpropagation)

XOR
Problem

ADALINE

Perceptron

Electronic Brain

1943

1957

1960

1969

1986

1995

2000

1940

1950

1960

1970

1980

1990

2000

20

History of machine learning.

Golden Age

Dark Age ("AI Winter")



McCulloch - W. Pitts



F. Rosenblatt



B. Widrow - M. Hoff



M. Minsky - S. Papert



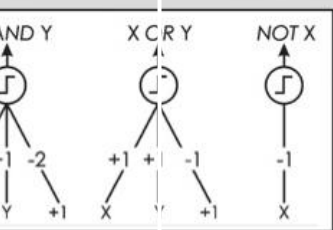
D. Rumelhart - G. Hinton - R. Williams



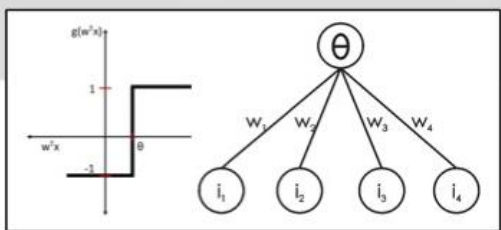
V. Vapnik - C. Cortes



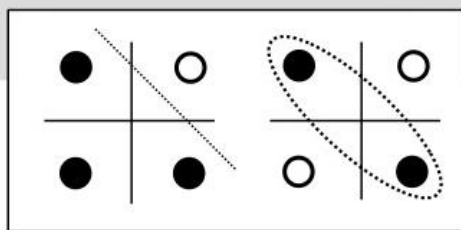
G. Hinton



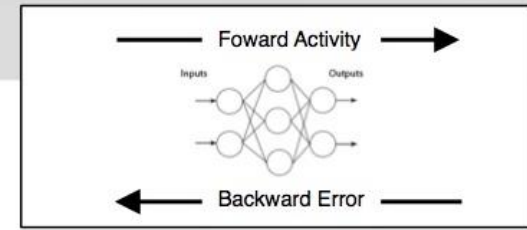
Adjustable Weights
Weights are not Learned



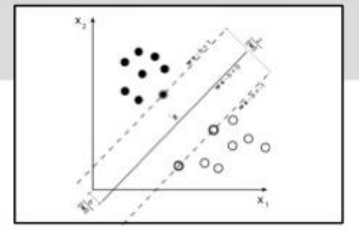
• Learnable Weights and Threshold



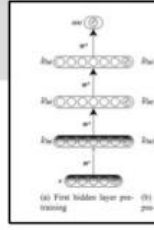
• XOR Problem



• Solution to nonlinearly separable problems
• Big computation, local optima and overfitting

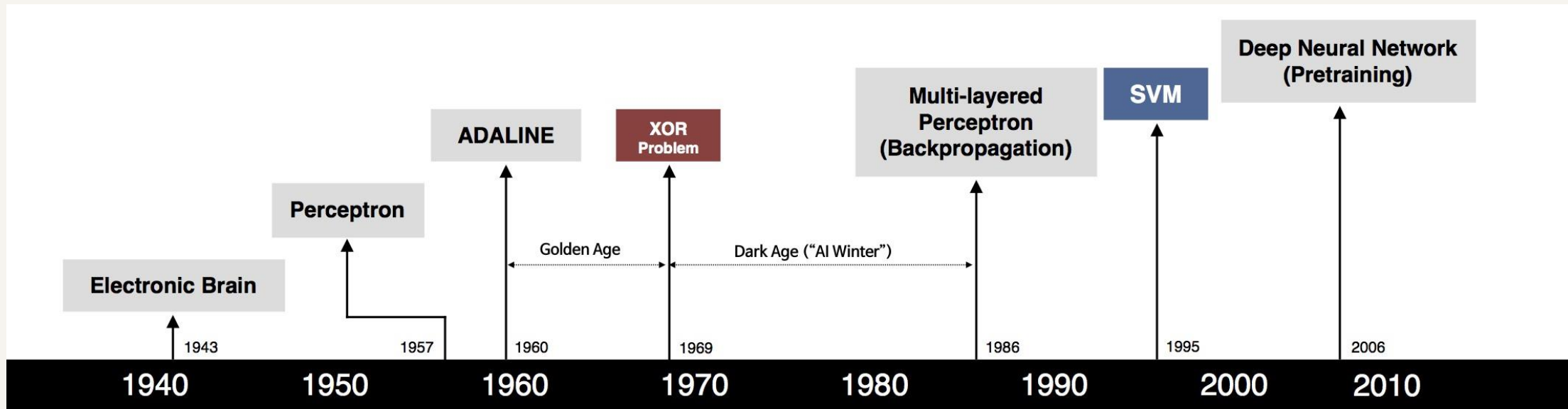


• Limitations of learning prior knowledge
• Kernel function: Human Intervention

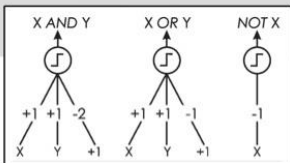


• Hierarchical

History of machine learning.



S. McCulloch - W. Pitts



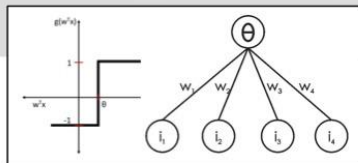
- Adjustable Weights
- Weights are not Learned



F. Rosenblatt



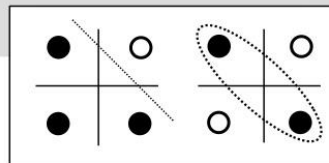
B. Widrow - M. Hoff



- Learnable Weights and Threshold



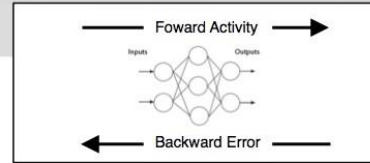
M. Minsky - S. Papert



- XOR Problem



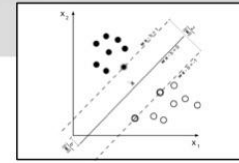
D. Rumelhart - G. Hinton - R. Williams



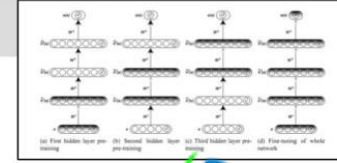
- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting
- Limitations of learning prior knowledge
- Kernel function: Human Intervention



V. Vapnik - C. Cortes

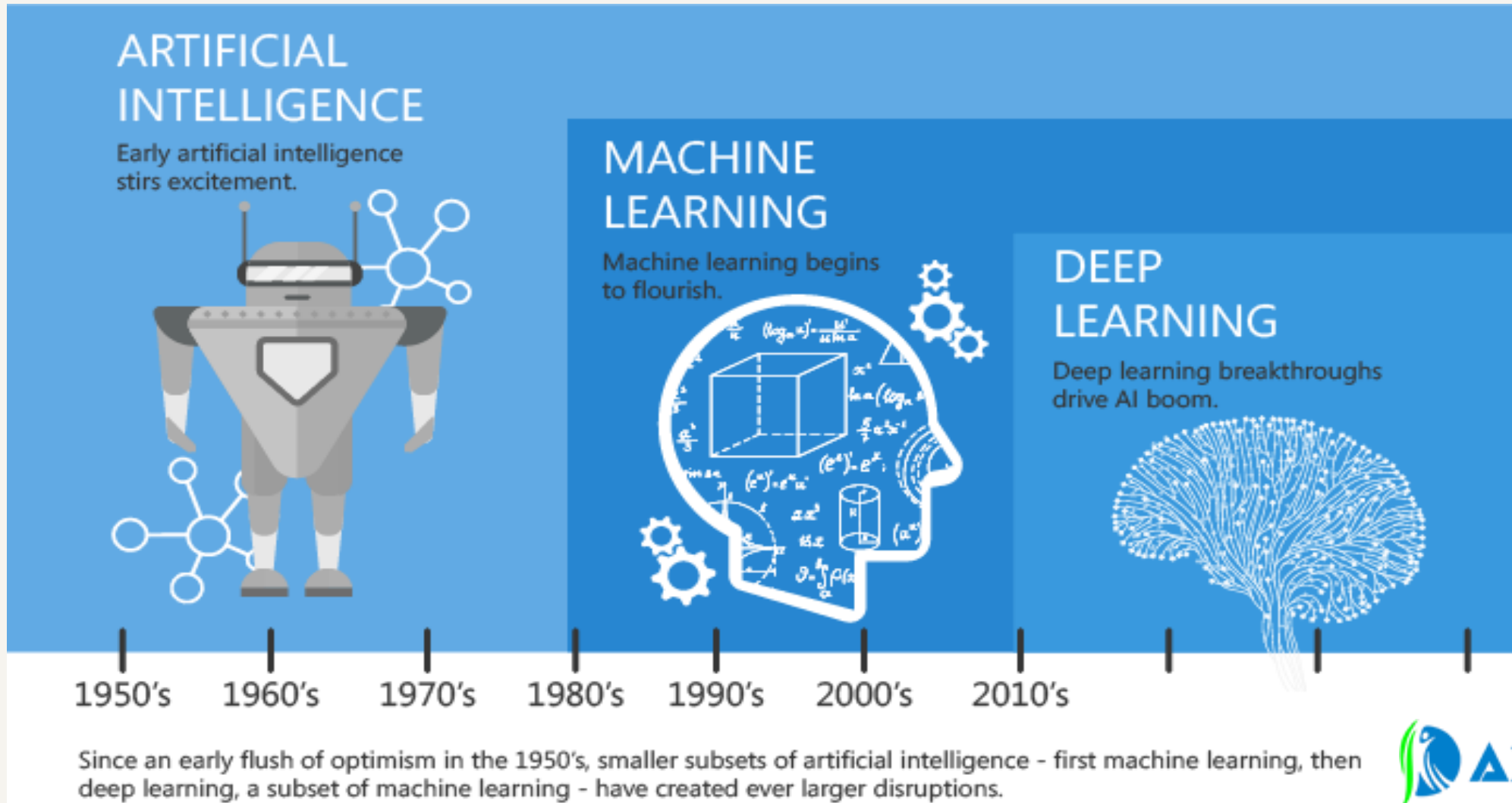


G. Hinton - S. Ruslan



- Hierarchical feature learning

History of machine learning.



History of machine learning.

- 1952 — Arthur Samuel wrote the first computer learning program. The program was the game of checkers, and the **IBM** computer improved at the game the more it played, studying which moves made up winning strategies and incorporating those moves into its program.
- 1957 — Frank Rosenblatt designed the first neural network for computers (the perceptron), which simulate the thought processes of the human brain.
- 1967 — The “nearest neighbor” algorithm was written, allowing computers to begin using very basic pattern recognition. This could be used to map a route for traveling salesmen, starting at a random city but ensuring they visit all cities during a short tour.
- 1979 — Students at Stanford University invent the “Stanford Cart” which can navigate obstacles in a room on its own.
- 1981 — Gerald Dejong introduces the concept of Explanation Based Learning (EBL), in which a computer analyses training data and creates a general rule it can follow by discarding unimportant data.

History of machine learning.

- 1985 — Terry Sejnowski invents NetTalk, which learns to pronounce words the same way a baby does.
- 1990s — Work on machine learning shifts from a knowledge-driven approach to a data-driven approach. Scientists begin creating programs for computers to analyze large amounts of data and draw conclusions — or “learn” — from the results.
- 1997 — IBM’s Deep Blue beats the world champion at chess.
- 2006 — Geoffrey Hinton coins the term “deep learning” to explain new algorithms that let computers “see” and distinguish objects and text in images and videos.
- 2010 — The [Microsoft](#) Kinect can track 20 human features at a rate of 30 times per second, allowing people to interact with the computer via movements and gestures.
- 2011 — IBM’s Watson beats its human competitors at Jeopardy.

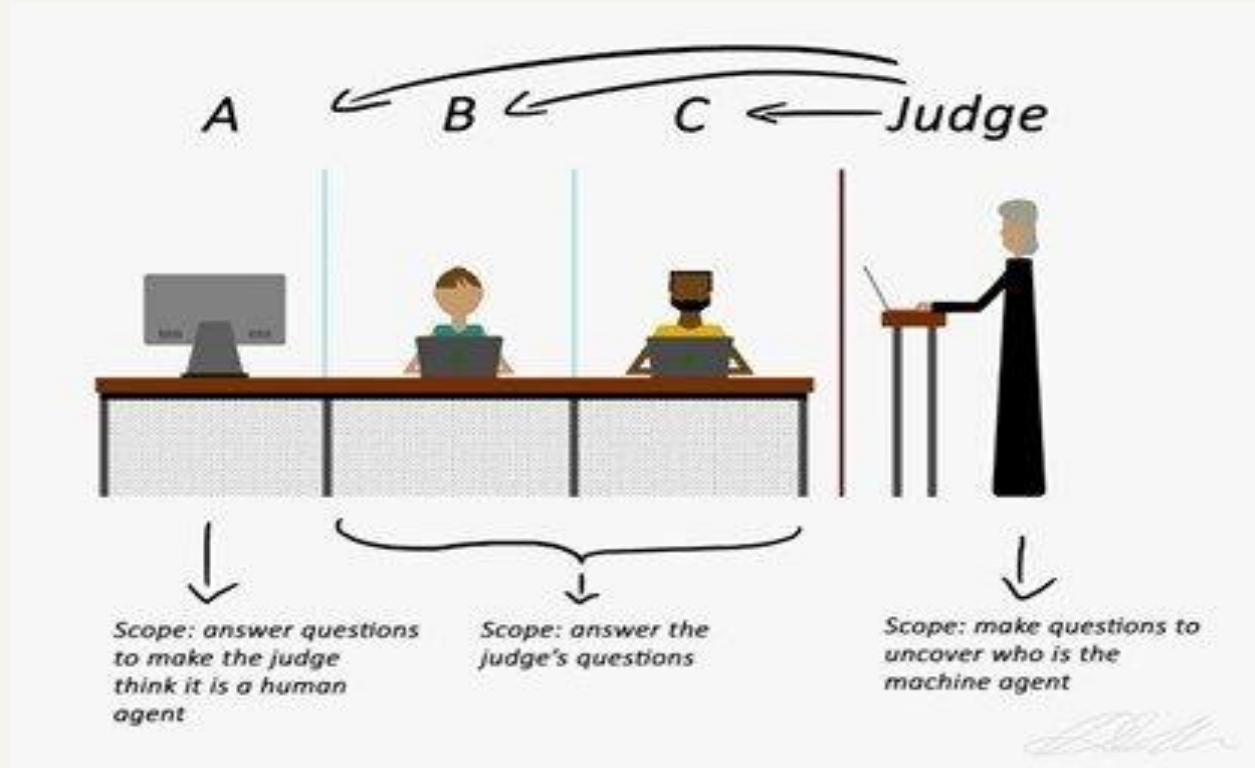
History of machine learning.

- 2011 — [Google](#) Brain is developed, and its deep neural network can learn to discover and categorize objects much the way a cat does.
- 2012 – Google’s X Lab develops a machine learning algorithm that is able to autonomously browse YouTube videos to identify the videos that contain cats.
- 2014 – [Facebook](#) develops DeepFace, a software algorithm that is able to recognize or verify individuals on photos to the same level as humans can.
- 2015 – Amazon launches its own machine learning platform.
- 2015 – Microsoft creates the Distributed Machine Learning Toolkit, which enables the efficient distribution of machine learning problems across multiple computers.

History of machine learning.

- 2015 – Over 3,000 AI and Robotics researchers, endorsed by Stephen Hawking, [Elon Musk](#) and Steve Wozniak (among many others), sign an open letter warning of the danger of autonomous weapons which select and engage targets without human intervention.
- 2016 – Google’s artificial intelligence algorithm beats a professional player at the Chinese board game Go, which is considered the world’s most complex board game and is many times harder than chess. The AlphaGo algorithm developed by Google DeepMind managed to win five games out of five in the Go competition.
- Self driving car by tesla
- Continue.....

Very first use case of machine learning.



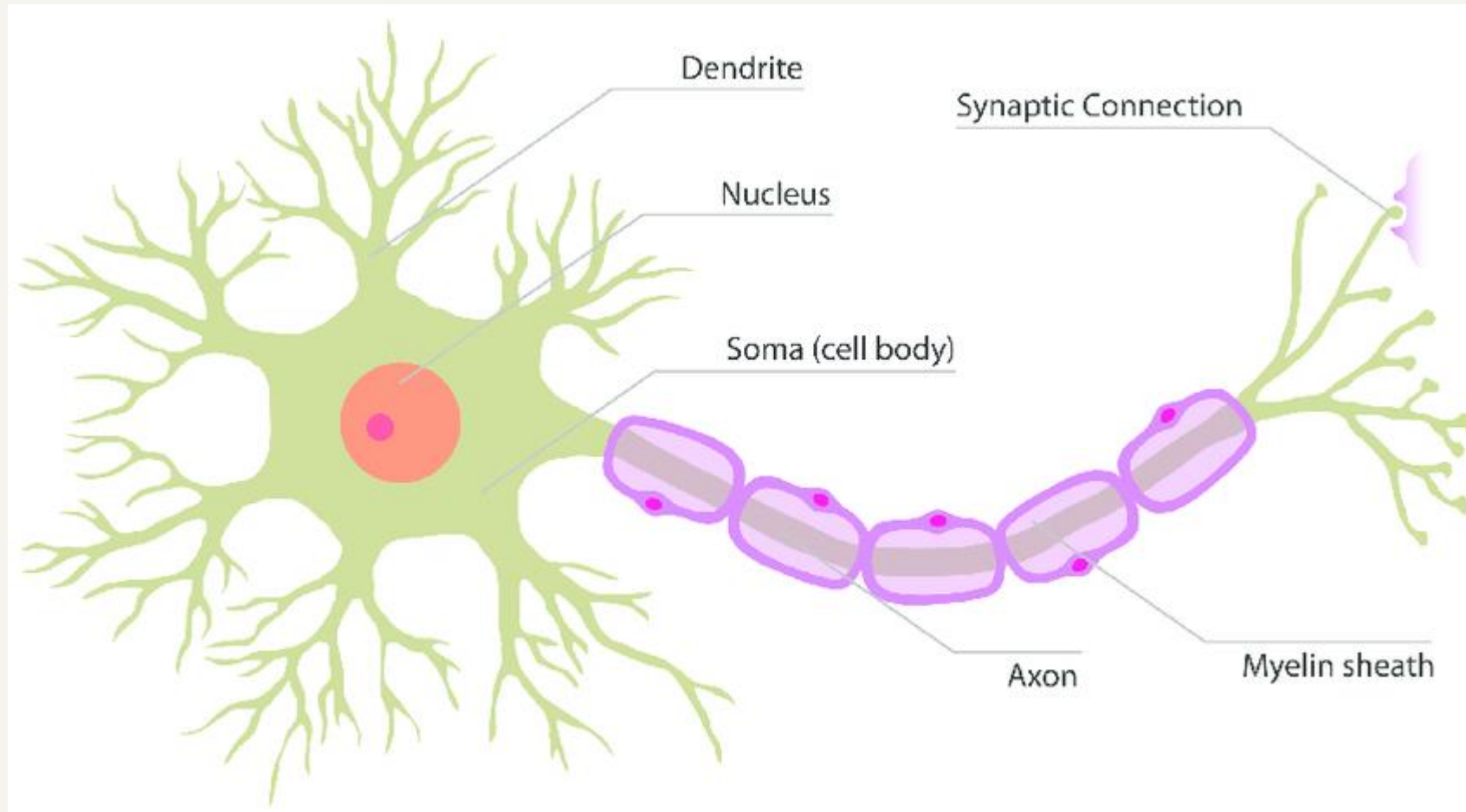
Very first use case of machine learning.

- In 1950 Alan Turing tried to imagine a test to decide whether a machine is intelligent.
- The test was constructed so a human judge would ask questions to multiple participants and had to discover which of the participants was a machine.
- To do so the judge is allowed to pose questions and try to unmask the robot interlocutor from the answers.
- To pass the test machine must not be uncovered to be a computer.

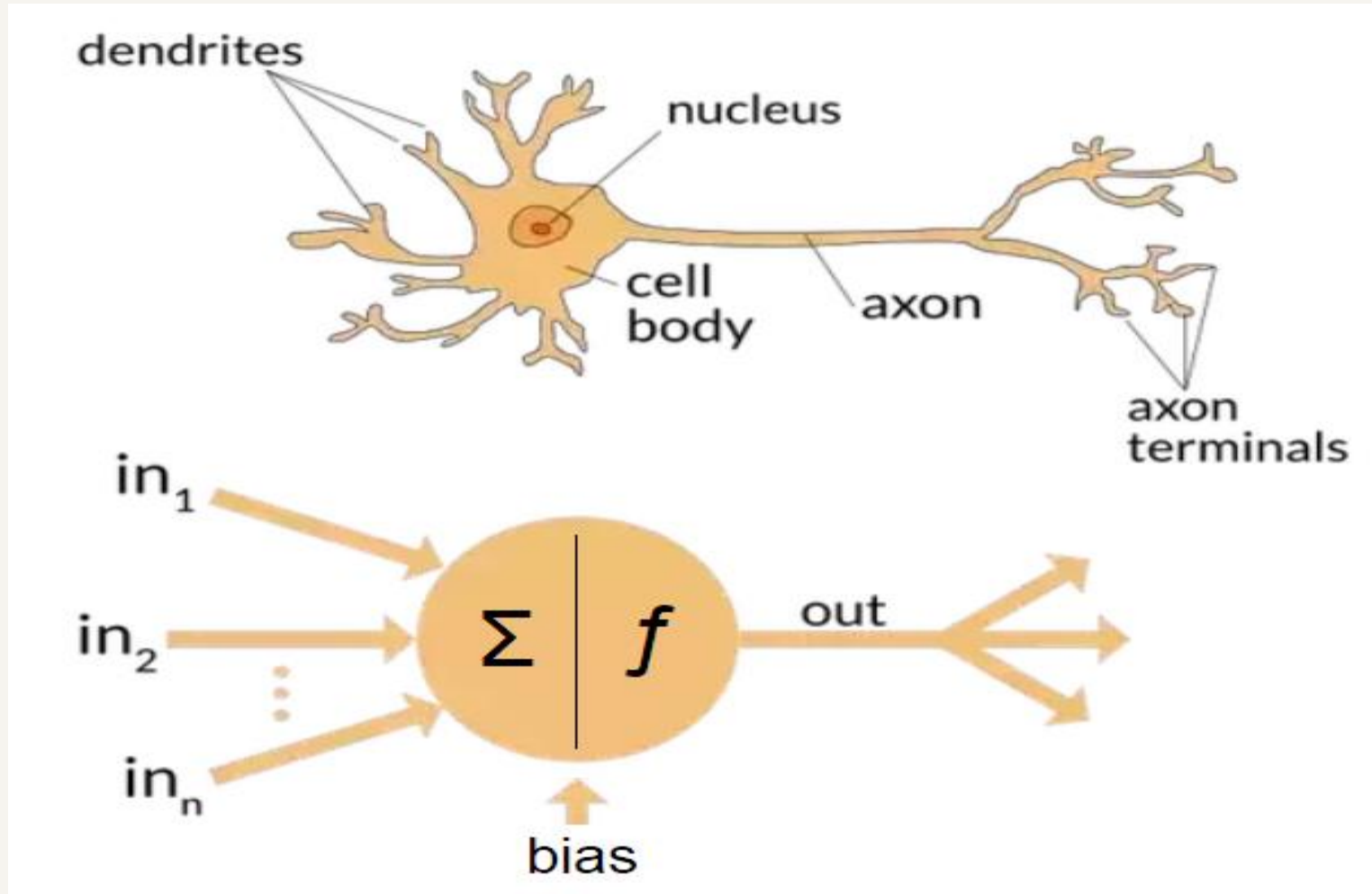
Very first use case of machine learning.

- Surprisingly (or not) very few AIs were able to pass the Turing test.
- The first was Eliza. Eliza was quite a simple program designed to mimic a psychologist. It takes the judges' questions reversing them and encouraging others to talk rather than properly answer the questions.
- Apart from Eliza a few other AIs have passed the Turing test.
- Most of them, instead of really being intelligent and have a natural conversation they use tricks to fool the judge.
- Other examples are Perry who imitates symptoms of schizophrenia by talking about itself.
- More Recently another AI poses as Eugene a 13 years old Ukrainian boy in an attempt to conceal grammatical incorrectness and misunderstanding.

Human Brain VS Machine

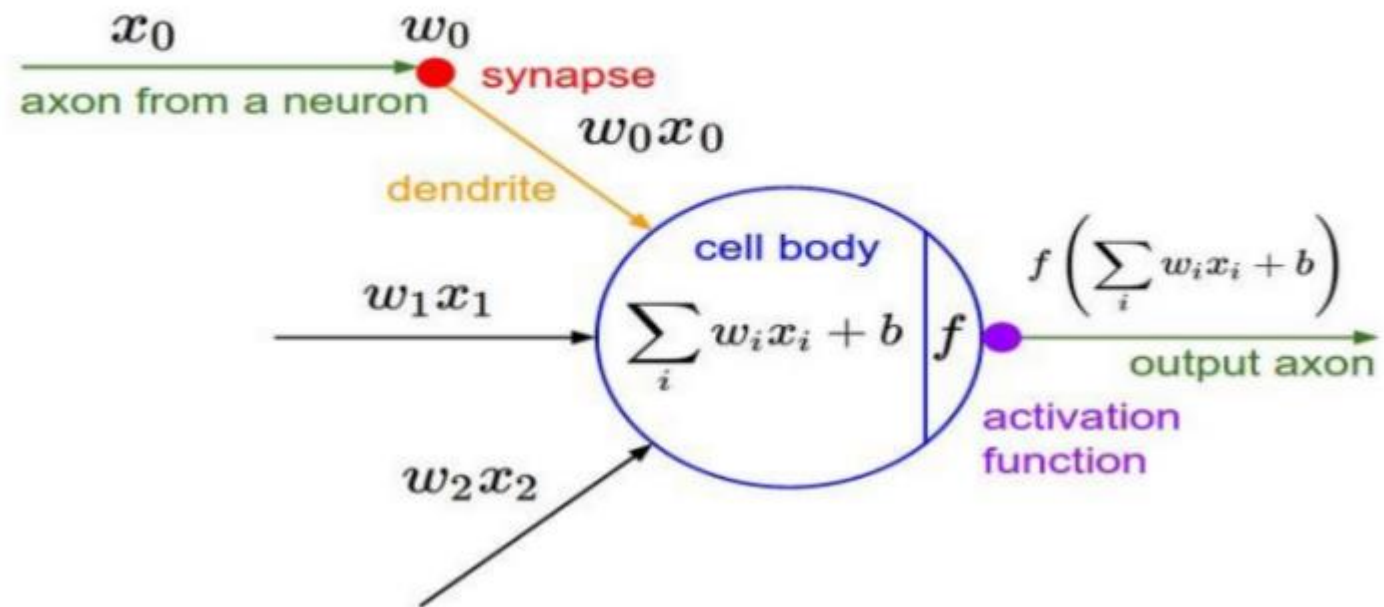


Human Brain VS Machine



Human Brain VS Machine

Artificial Neuron



Human Brain VS Machine

Brain Differences

now wait before you start thinking that you can just create a huge neural network and call strong AI, there are some few points to remember:

Just a list:

- The artificial neuron fires totally different than the brain
- A human brain has 100 billion neurons and 100 trillion connections (synapses) and operates on 20 watts(enough to run a dim light bulb) - in comparison the biggest neural network have 10 million neurons and 1 billion connections on 16,000 CPUs (about 3 million watts)
- The brain is limited to 5 types of input data from the 5 senses.
- Children do not learn what a cow is by reviewing 100,000 pictures labelled "cow" and "not cow", but this is how machine learning works.
- Probably we don't learn by calculating the partial derivative of each neuron related to our initial concept. (By the way we don't know how we learn)

Chapter 1: Introduction of Machine Learning



Thank You
Vishal Modi